

Vasily Dunaev

vasilysent@gmail.com | [Website](#) | [GitHub](#) | [LinkedIn](#) | Amsterdam, The Netherlands

Robotics R&D engineer with an MSc in Applied Mathematics and BSc in Robotics, focused on robot dynamics, control, numerical simulation, and mathematical modelling of physical systems. I work across geometric mechanics, simulation, and model-based and learning-based control, with hands-on experience building robotic systems.

SKILLS

Robot kinematics and dynamics, model-based control, numerical simulation, reinforcement learning;
Differential geometry, Lie groups and Lie algebras, geometric mechanics, optimisation;
Python, C++, Julia, MATLAB/Simulink, MuJoCo, Git, CAD, LaTeX.

EDUCATION

- **University of Twente, MSc in Applied Mathematics** Sep 2023 – Aug 2025
Enschede, The Netherlands
Grade: 9.8/10
Thesis: Geometric structure of Cosserat elasticity [1]. Supervisor: [Frederic P. Schuller](#).
- **University of Twente, Erasmus+ Exchange** Feb 2022 – Feb 2023
Enschede, The Netherlands
Grade: 9/10
- **ITMO University, BSc in Robotics** Sep 2019 – Jun 2023
St Petersburg, Russia
Grade: 4.6/5
Designed and organised an experimental advanced BSc programme in Robotics focused on fundamental subjects. Initially created for my group, the programme was later adopted for subsequent cohorts.

EXPERIENCE

- **Reinforcement Learning Robotics Project** May 2026
Built a MuJoCo simulation of a torque-controlled UR5e for 6D end-effector pose tracking. Combined a geometric SE(3) spring-damper controller with an RL residual policy that outputs a virtual end-effector wrench. [\[repo\]](#)
- **Graduate Researcher (University of Twente)** Sep 2025 – Sep 2026
Developed a differential-geometric formulation of the kinematics and dynamics of Cosserat continua, enabling the treatment of general continua with residual stresses [1]. Manuscript in preparation for submission, draft on request.
- **Instructor (University of Twente)** Sep 2022 – Nov 2025
Co-taught a master's course on robot modelling, kinematics, and dynamics with [Stefano Stramigioli](#). Covered rigid transformations, twists, wrenches, robot kinematics, dynamics, and torque control. Delivered lectures and tutorial sessions; designed assignments and exams; produced lecture notes [2] and [video lectures](#).
- **Research Intern (KdVI Institute, University of Amsterdam)** Mar 2025 – Jun 2025
Built reusable software for structure-preserving numerical integration on matrix Lie groups, including stable simulation tools for rigid-body dynamics on SO(3) and SE(3). [\[report\]](#)
- **Engineer (Youth Robotics Lab, ITMO University)** Sep 2016 – Jan 2022
Selected activities:
 - **Robot-Artist**
Led a team of eight building a painting robot. Implemented robot's motion control, stroke-sequence generation using genetic and simulated-annealing algorithms, designed and built the hardware. [\[photos and videos\]](#)
 - **ELSA**
Created a humanoid robot that plays the flute. In particular, designed the CAD model and built the 3D-printed robot's hands. RoboCup Junior 2019 winner. [\[video in Russian, English CC available\]](#)
 - **Strawberry Fields**
Developed an autonomous strawberry-harvesting prototype; trained a CNN for berry detection and designed a soft robotic gripper. Won 1st place at the World Robot Olympiad 2018. [\[video\]](#)
 - **Teaching**
Developed and taught a course CAD 3D modelling and 3D printing.

PUBLICATIONS

- [1] V. Dunaev, [“Geometric structure of Cosserat elasticity,”](#) Thesis, University of Twente, Enschede, Aug. 2025.
- [2] V. Dunaev, [Modelling, Dynamics, Kinematics. Lecture notes.](#) University of Twente, 2025.
- [3] A. A. Peregudin, A. M. Popov, E. R. Akhmetgaliev, A. D. Bugrova, M. A. Bystramovich, V. A. Stakanov, and **V. A. Dunaev**, [“Virtual laboratory for game-based control systems education*,”](#) *IFAC-PapersOnLine*, vol. 55, no. 17, pp. 344–349, 2022, 13th IFAC Symposium on Advances in Control Education ACE 2022, ISSN: 2405-8963.

SELECTED AWARDS

Recipient – University of Twente Scholarship Enschede, The Netherlands Value: €12,000 per year	2023 – 2025
1st and 2nd Places – RoboCup Sydney, Australia Junior OnStage category	2019
1st Place – World Robot Olympiad Chiang Mai, Thailand Open category (Senior)	2018
1st Place – RobotChallenge Beijing, China Freestyle category	2018